

Resolving Quantity and Informativeness Implicature in Indefinite Reference

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July 18

XPRAG 2015

The Phenomenon



The man injured his child.

The man injured someone else's child.

The man injured a child.

The Phenomenon



The man injured his child.

The man injured someone else's child.

The man injured a child.

The Phenomenon...

...and the research question

The X V-ed a Y.

What determines this
variation in the
directionality and **strength**
of inferences about
utterance meaning?



The man injured his child.

The man broke a finger.

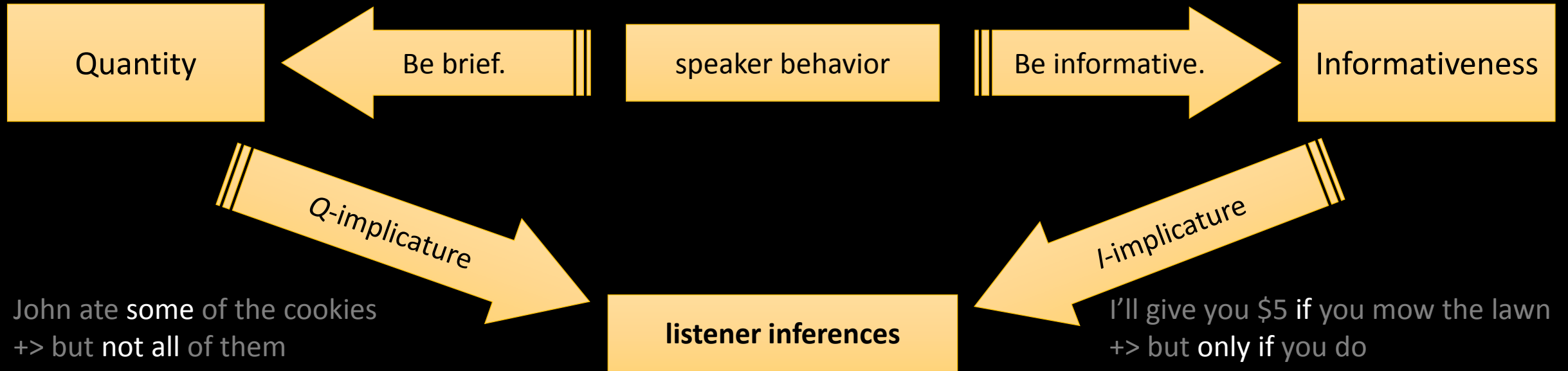
The man broke a nose.

The man injured a child.

The man injured a son.

The man injured someone else's child.

Gricean inferences



The *X* V-ed a *Y*.

Grice (1967, 1975); Zipf (1949), (1967, 1985), Zipf (1989), Levinson & Zipf (2009), Zipf (2011), Benci et al. (2014), Zipf (2016), Blizard (2016), Zipf (2019).

The Rational Speech Act (RSA) model

Assumption 1:
The “Lexicon”

The man injured a child.

The man injured his child.

The man injured someone else’s child.

Frank & Goodman (2012)

The Rational Speech Act (RSA) model

Assumption 1:
The “Lexicon”

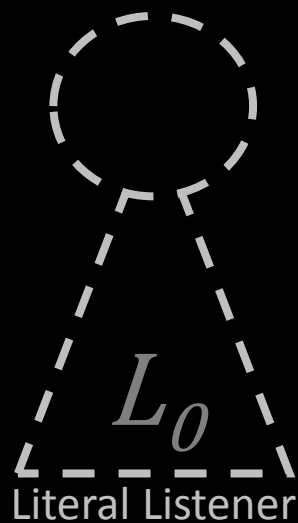
	OWN	OTHER'S
The man injured a child.	1	1
The man injured his child.	1	0
The man injured someone else's child.	0	1

Assumption 2:
Utterance costs

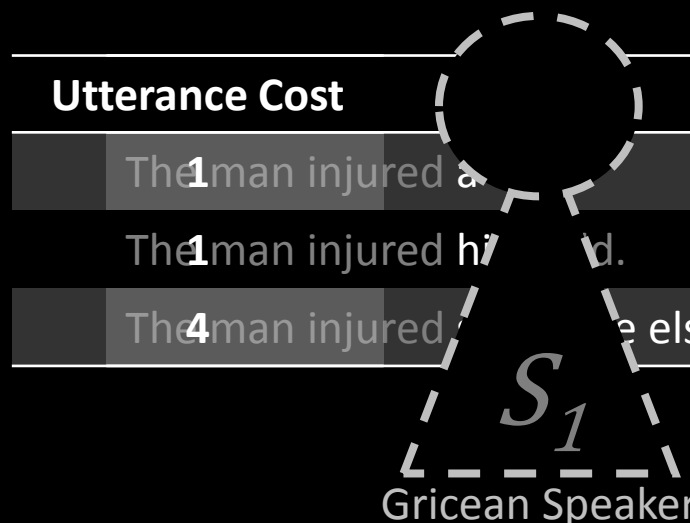
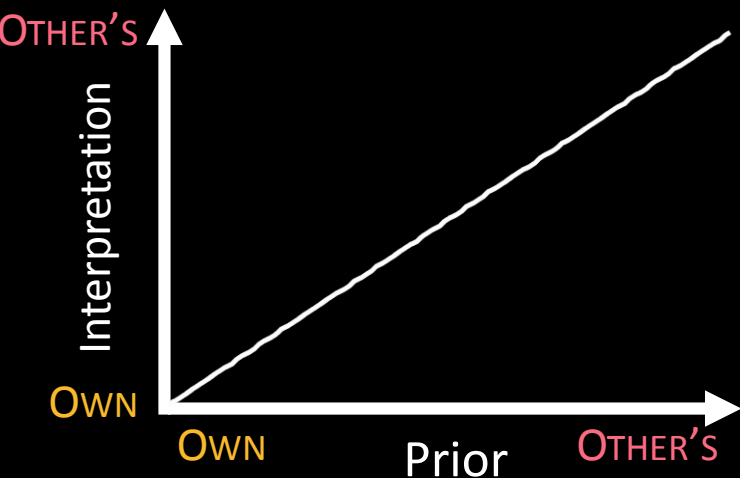
$D(a) =$	1
$D(his) =$	1
$D(someone\ else's) =$	4

Frank & Goodman (2012)

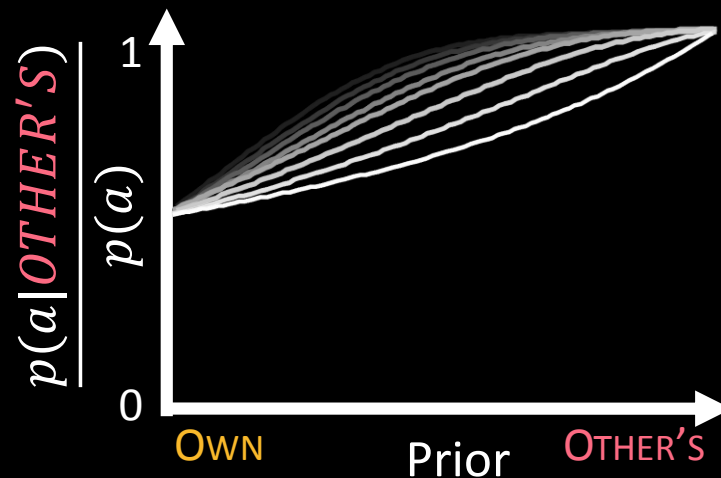
The Rational Speech Act (RSA) model



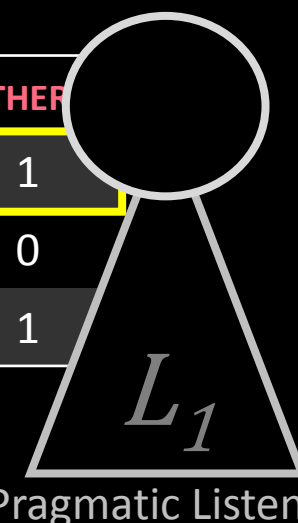
$$L_0 \sim \text{lexicon} * \text{prior}$$



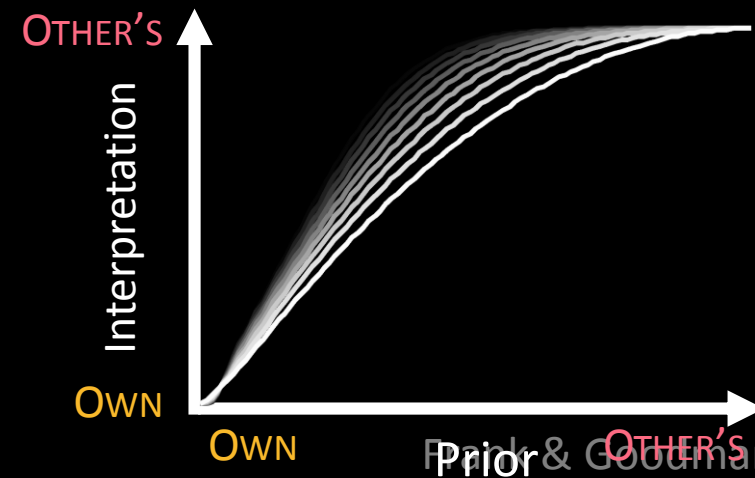
$$S_1 \sim \exp(\lambda \log(\log(L_0 / \text{cost})))$$



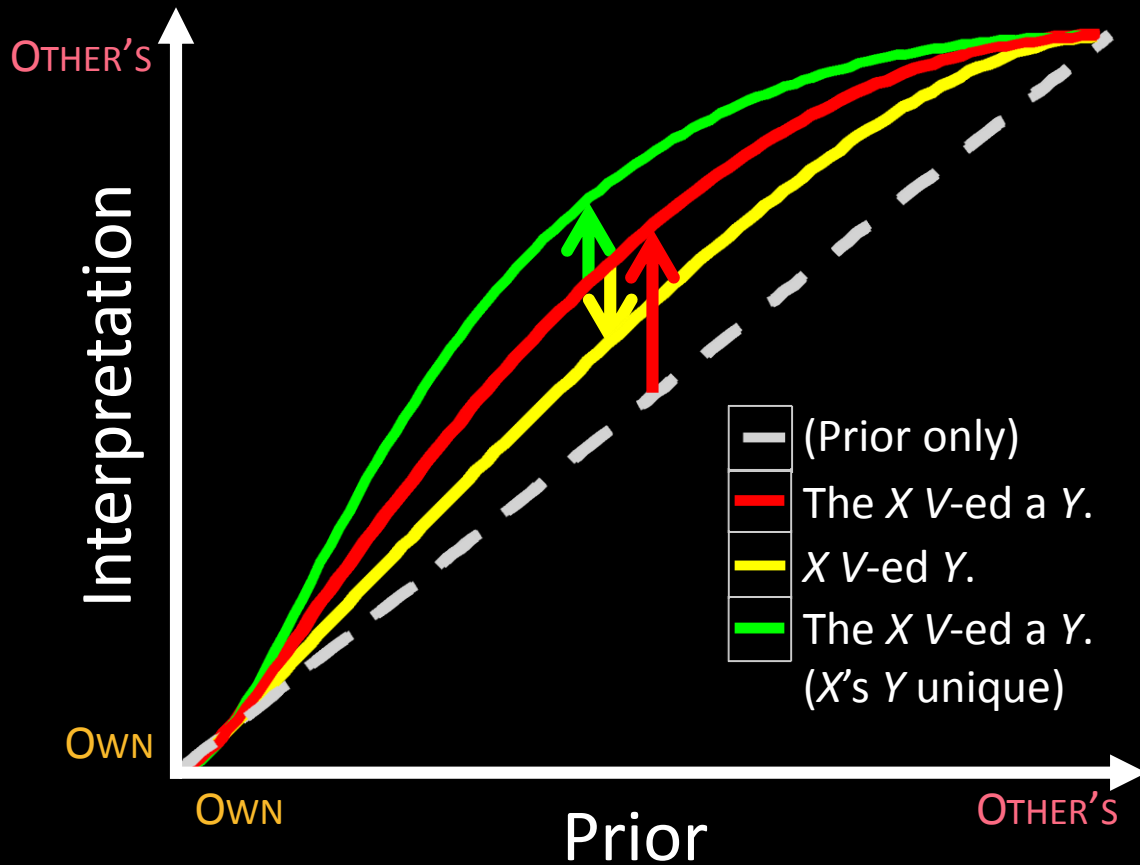
Utterance Cost		OWN	OTHER
The 1 man injured a		1	1
The 1 man injured his	d.	1	0
The 4 man injured	the else's child.	0	1



$$L_1 \sim \text{prior} * S_1$$



Predictions



1. Interpretations track priors
2. Baseline Q-implicature towards OTHER'S
3. Reduced Q-implicature in "headlines"

4. Strengthened Q-implicature

where X 's Y is unique

4 US president

someone else's

4

$D(a, \text{OWN}) > D(a, \text{OTHER'S})$

Frank & Good (2012), Frank (2012), Frank (2012), Frank (2012)

Methodology

Experiment 1: Interpretations

What do you think this sentence is about:

Man shaved leg.

- ☐ his own leg
- ☐ someone else's leg

Norming experiment: Priors

How likely is a man to save his own family compared to saving someone else's family?

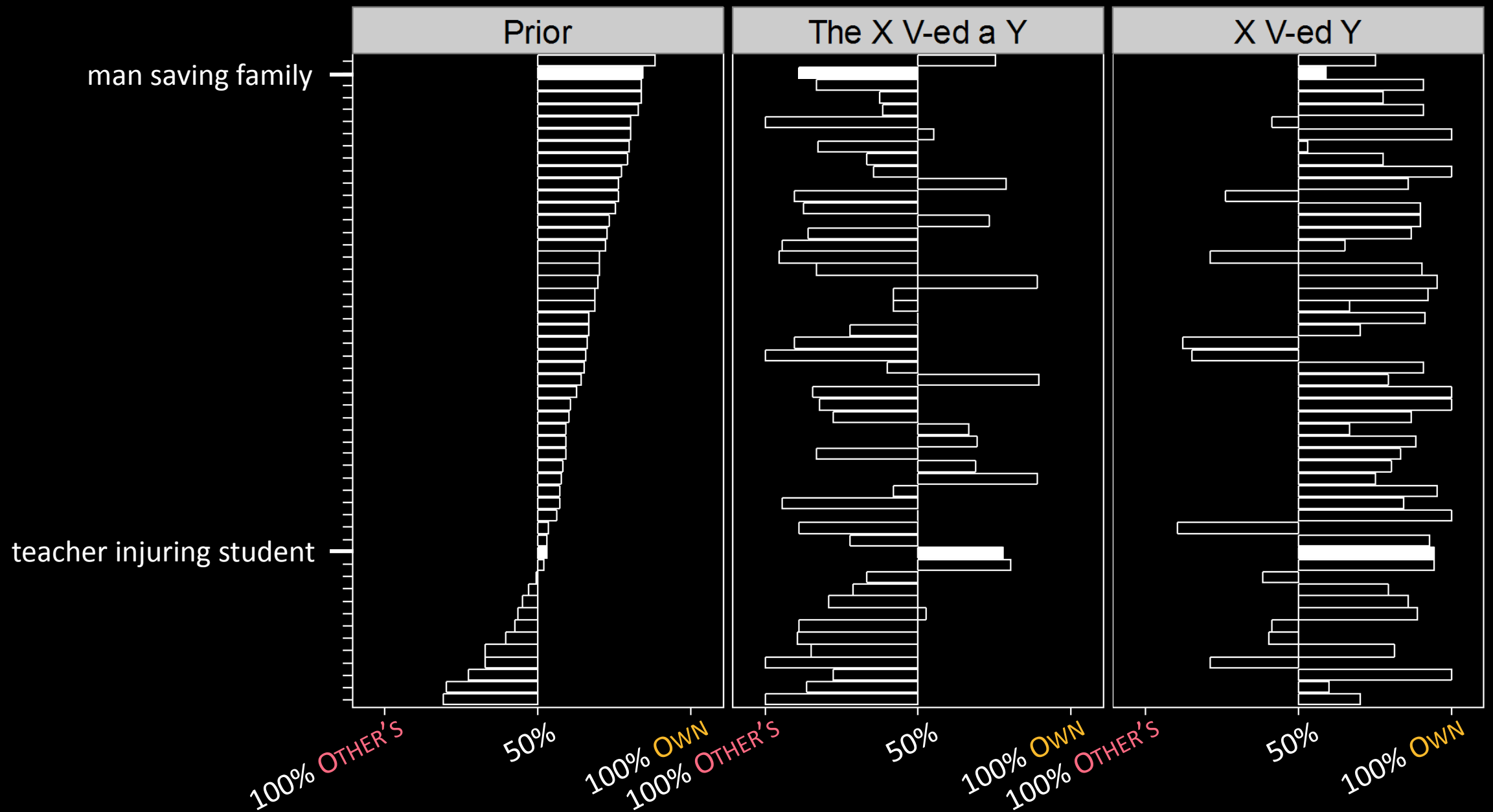
100% likely to save
his own family

equally likely

100% likely to save
someone else's
family

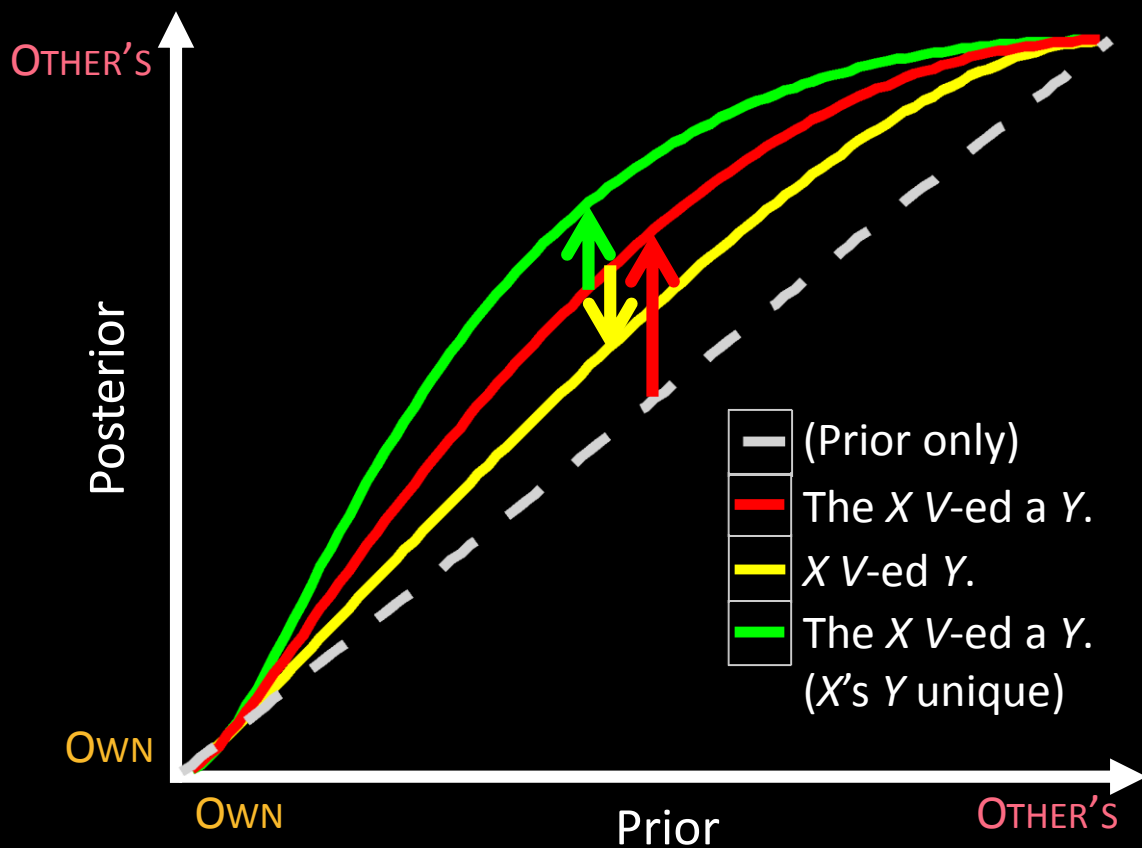


$\text{response} \sim \text{prior} + \text{XUniqueness} + \text{reliability} + \text{headline} + (1 + \text{headline} \mid \text{item})$

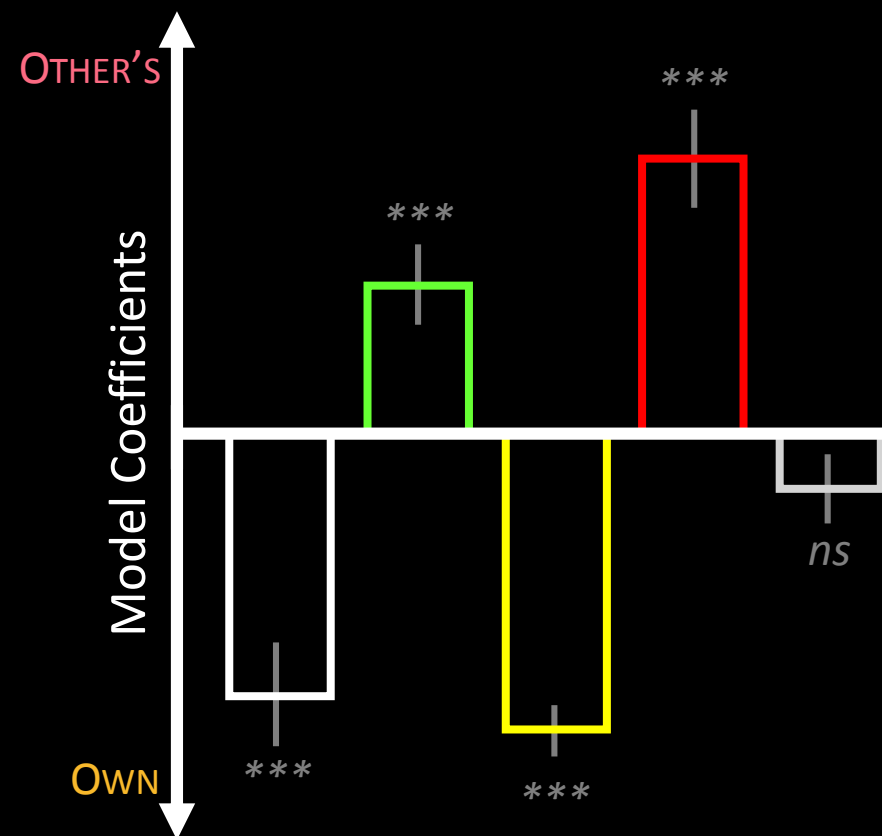


Results

Model Predictions



Regression Results



Discussion point 1 of 3: No effect of the prior?

3 possibilities:

1. Noisy measures

2. Maybe RSA got it wrong?
Norming experiment: Priors

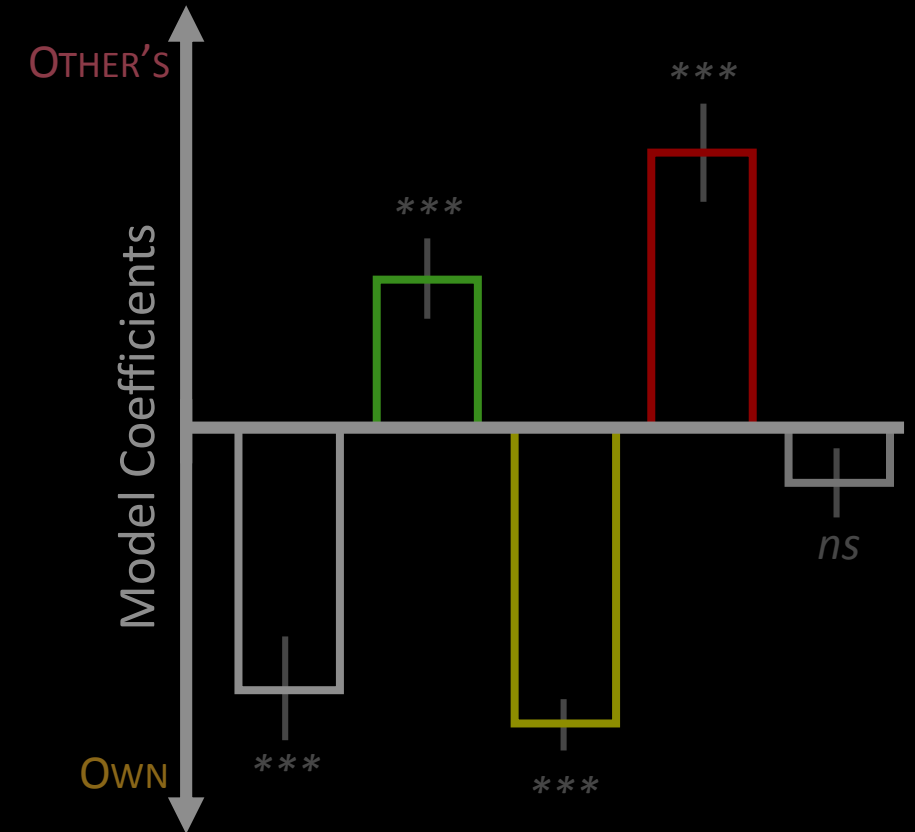
3. Event priors vs “intention priors”

How likely is a man to save his own family compared to saving someone else's family?

100% likely to save
his own family

equally likely

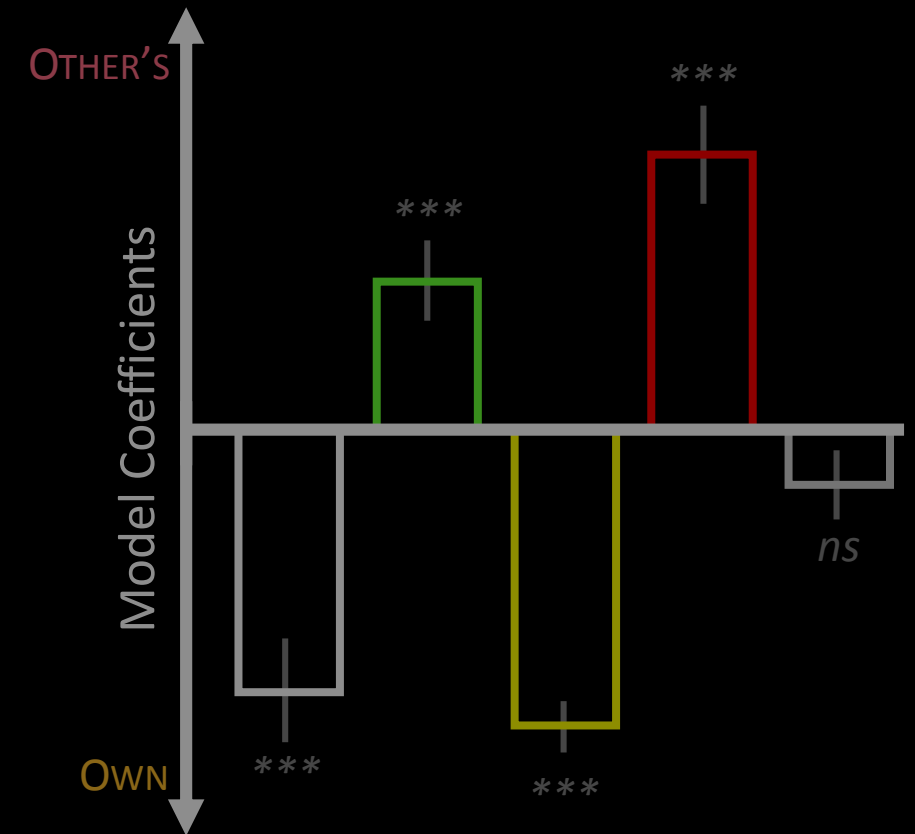
100% likely to save
someone else's
family



Discussion point 2 of 3: Support for RSA

RSA excels at predicting Q-implicatures:

1. Overall OTHER'S skew
2. Opposing trend in headline versions
3. Enhanced Q-implicature where *X*'s *Y* is unique



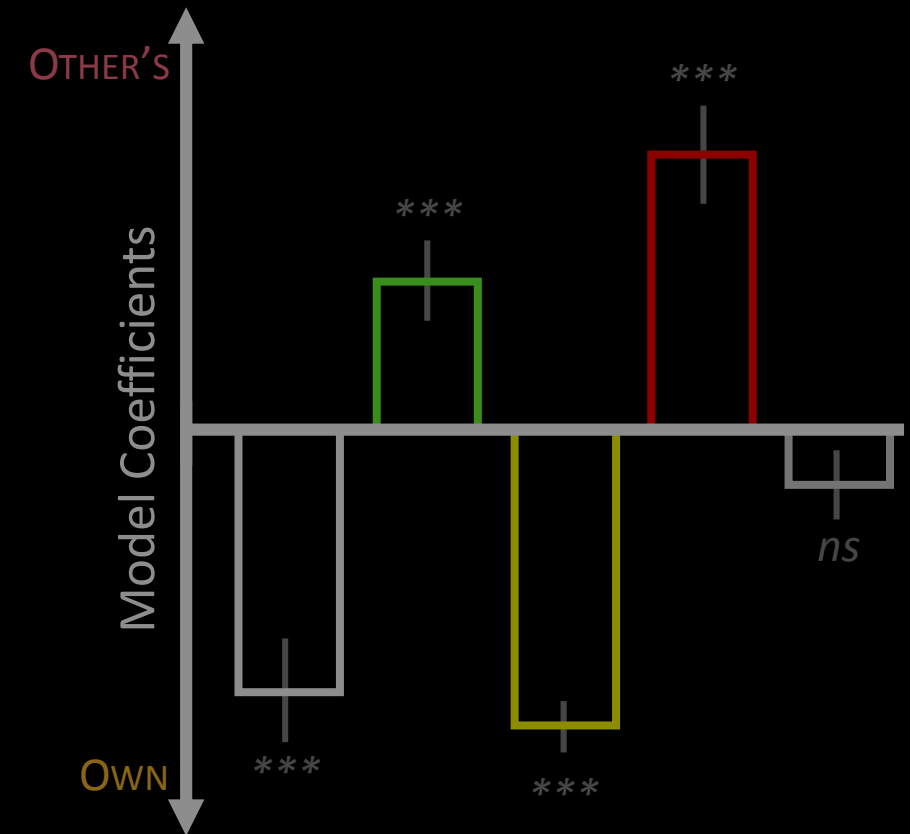
Discussion point 3 of 3: Relatability – Q or I (or neither?)

The X V-ed a Y.

The man injured a child.
The father injured a child.

2 possible reasons:

- *Ad hoc* Q-implicature about referring expressions (e.g. *man* vs. *father*)
- I-driven inference from real-world knowledge about the event participants (cf. I almost bought a car today but the engine was too noisy.)



Hirschberg (1985); Clark (1975); Prince & Cole (1981); see also Cohen & Kehler (in prep)

Conclusion

What we have learned

- Forced-choice experiments and mixed-logit models: great for studying interpretational preferences
- *Q/I* resolution is determined by multiple interacting factors
- RSA captures the essence of *Q*-implicature
- We don't understand *Informativeness* nearly as well
- Inference taxonomies may become explanatorily obsolete

Where to go from here

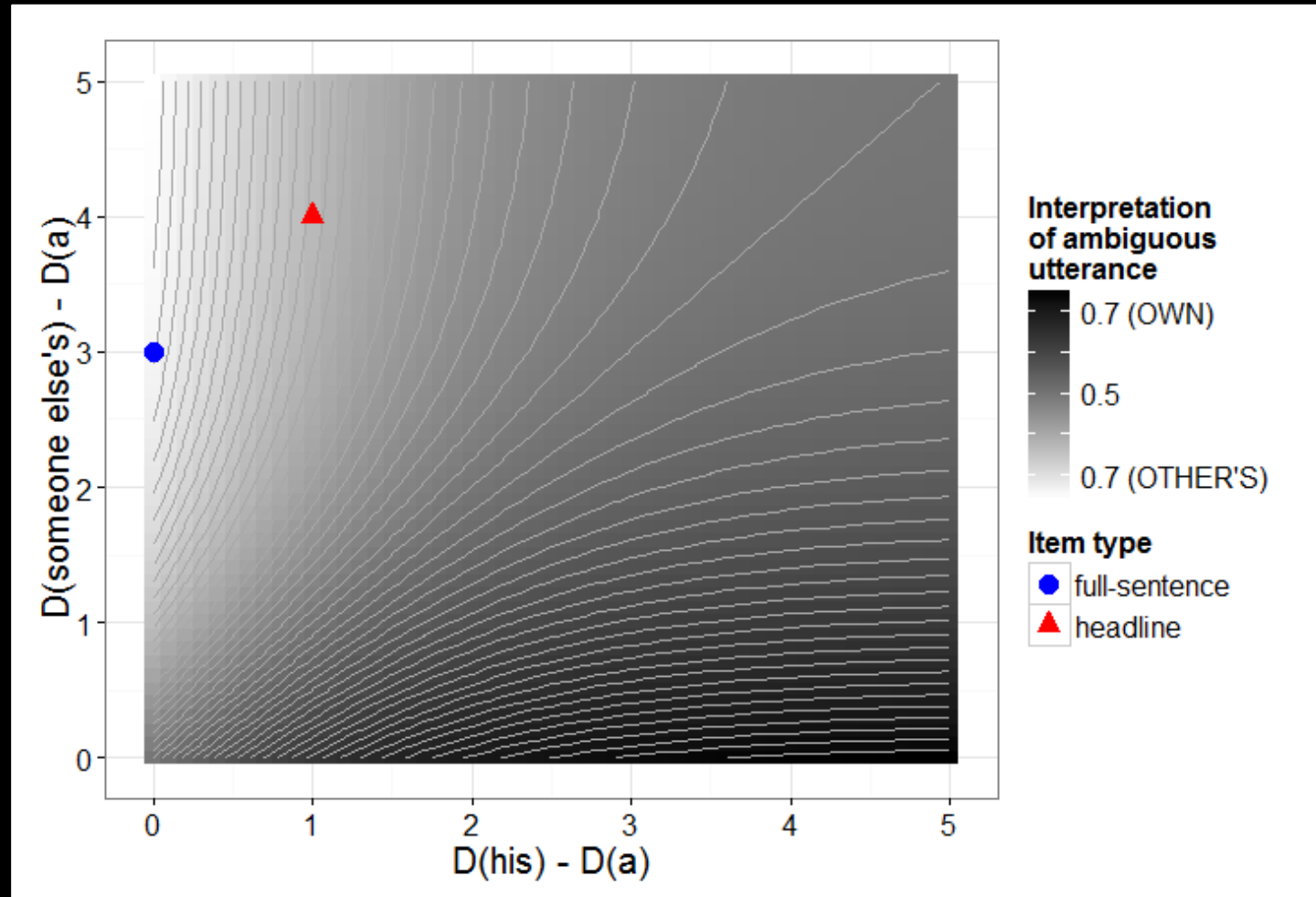
- Cross-linguistic validation of RSA
- More research on *I*-driven inferences

Thank you.

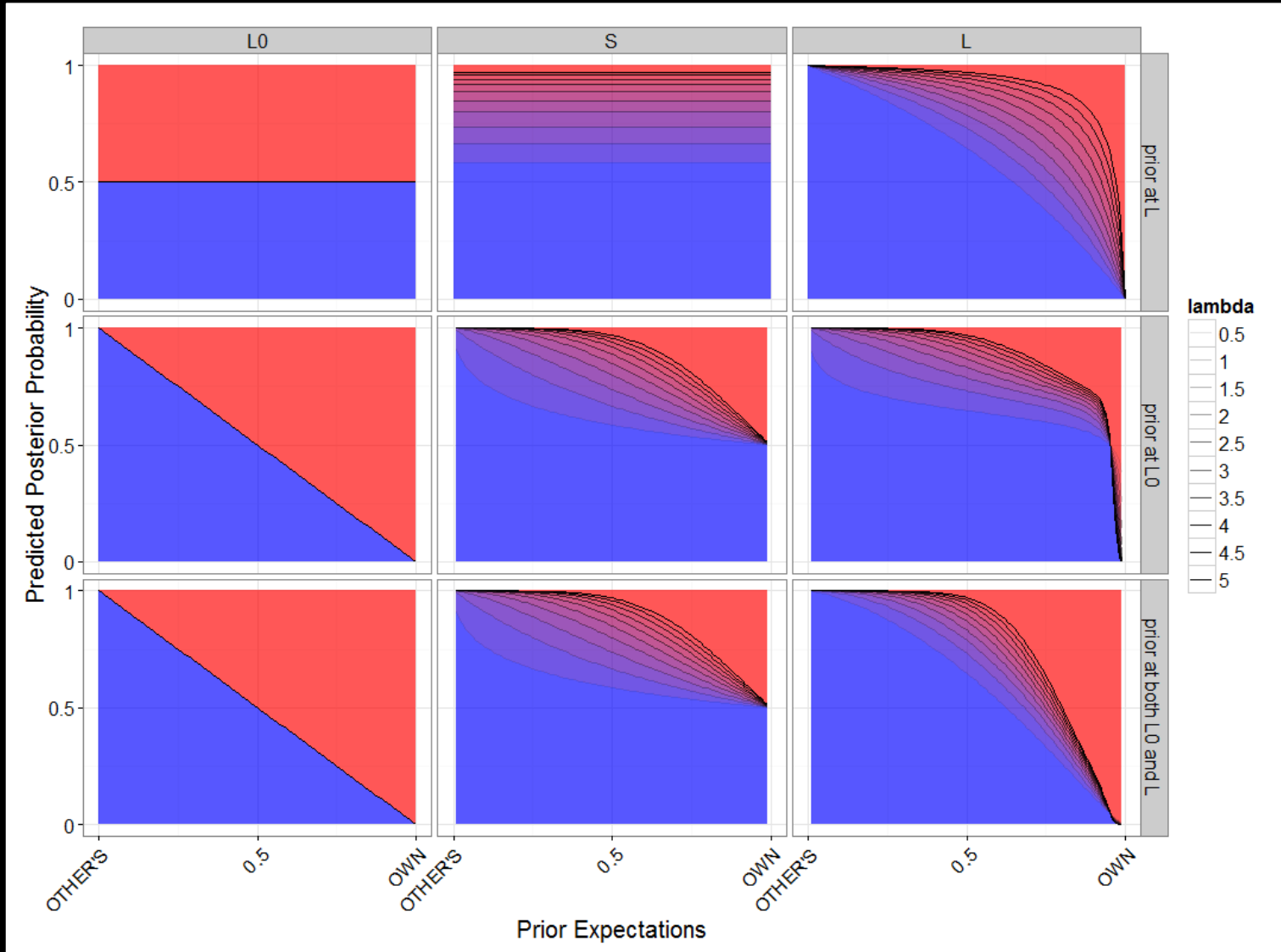
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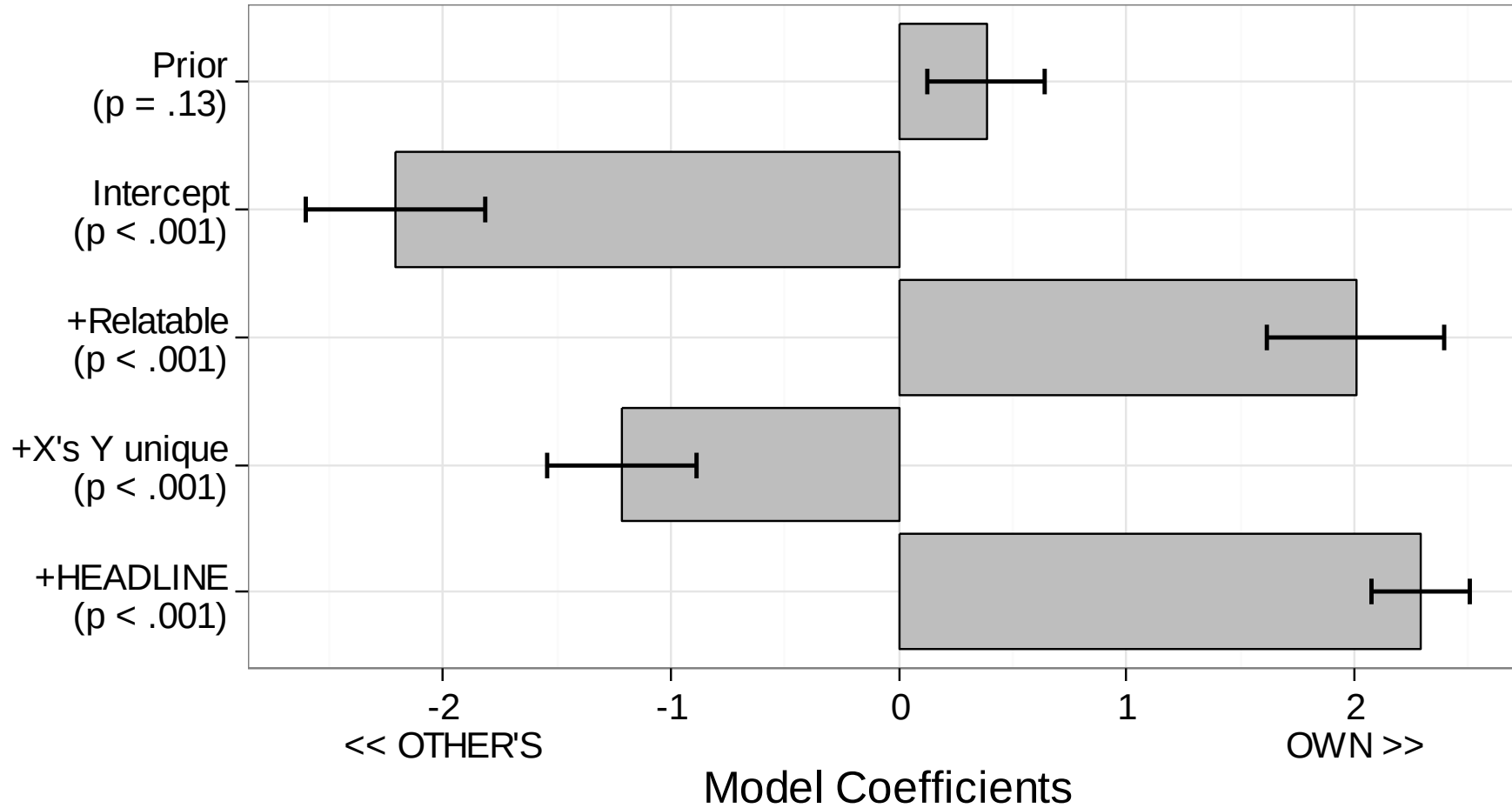
RSA predictions by disambiguation costs



Comparing RSA implementations



Regression results



Logit transform

